

Assignment (08 - 07 - 26)

1. What does API stand for, and what is its primary purpose in software communication?

API stands for Application Programming Interface. Its primary purpose is to act as a bridge that allows two different software applications to communicate with each other.

2. Using the restaurant analogy from the session, explain the role of the "waiter" in an API request.

In the restaurant analogy, the customer is the client, the kitchen is the server, and the API is the waiter. The waiter takes the customer's order (the request) to the kitchen and later brings the prepared dish (the response) back to the customer.

3. Give two real-world examples of applications that rely on APIs (other than those covered in the session) and name the type of API each likely uses.

Netflix uses a Video Streaming API to fetch and stream movies/shows to your device based on what you select.

PhonePe or Google Pay uses a Banking/UPI API to connect to your bank and complete money transfers securely.

4. Why do companies prefer using existing APIs (for maps, payments, login, etc.) instead of building these services from scratch?

Building things like maps or payment systems from scratch takes huge time, money, and expertise. Using existing APIs lets companies plug in ready-made, tested services instantly and focus on their own product instead.

5. Differentiate between the roles of a "Client" and a "Server" in client-server architecture, with one example of each.

The Client is the application that sends a request for a service or data (Example : a browser asks for customer records). The Server is the application that receives that request, processes it, and provides the requested service or data back as response. (Example: a database server storing customer records response back with the customers details)

6. What is the purpose of DNS, and why can't computers communicate using domain names alone?

DNS (Domain Name System) translates human-friendly domain names, like google.com, into the numeric IP address that computers actually use to locate each other on the network. Computers can't understand human language to route data, so DNS is needed.

7. Briefly explain how data physically travels across the internet in packets before reaching its destination.

Data is broken into small packets before sending. Each packet travels through the ISP and routers separately, and a DNS lookup finds the right server. Once all packets arrive, they are joined back together.

8. Define HTTP and explain why it is described as a "protocol."

HTTP (Hyper Text Transfer Protocol) is the set of rules used to send requests and responses between a browser and a server. It's called a protocol because it's an agreed set of rules, just like people use a common language to talk.

9. List the five components that make up an HTTP request.

URL – where to send the request

Headers – extra info like format or browser

Body – the actual data being sent

Authentication – proof of identity

Method – the action to perform (GET, POST, etc.)

10. What is the difference between `response.status_code` and `response.json()` when reading an API response?

`status_code` just gives the number showing success or failure (like 200 or 404).

`json()` gives the actual data from the response, converted into a Python dictionary.

11. Using the pizza-ordering analogy, match each part of an order (URL, Method, Headers, Body, Authentication) to its real-world equivalent.

URL – the pizza shop's address

Method – POST, since it's a new order

Headers – your address and phone number

Body – the order details (size, toppings)

Authentication – your account or phone number

12. Break down the URL <https://api.openai.com/v1/chat/completions> into its four components.

- 1) https:// – Protocol – secure connection method
- 2) api.openai.com – Domain – server address
- 3) v1 – Version – API version
- 4) chat/completions – Endpoint – the specific action requested

13. What is an API endpoint? Give an example of two different endpoints for a shopping application.

An endpoint is one specific task an API can do.

Example: /products to browse items & /cart to manage the cart.

14. Match each method to its purpose.

GET – read data (view employees)

POST – add new data (add employee)

PUT – replace the whole record (update all details)

PATCH – update one field (change salary only)

DELETE – remove data (delete employee)

15. Difference between PUT and PATCH?

PUT replaces the whole record – anything missing may get deleted. PUT for full replacements.

PATCH only updates the fields you send, leaving the rest unchanged. Use PATCH for small updates

16. What do 401 and 403 mean, and how are they different?

401 (Unauthorized) - means the request has no valid authentication at all – the server doesn't know who you are because the API key is missing or invalid.

403 (Forbidden) - means the server does recognize your identity, but you don't have permission to access that particular resource.

17. Why is 429 returned, and what should the app do?

429 means too many requests were sent too fast. The app should wait a bit and then retry, instead of sending requests again immediately.

18. Purpose of "Authorization: Bearer <API_KEY>"?

It proves who you are to the server, like showing an ID card, so the server knows the request is authorized.

19. Three golden rules for handling API keys.

Never hard-code keys in your code or upload them to GitHub.

Always use HTTPS.

Rotate keys often and limit what each key can access.

20. Why is JSON the standard format for API responses?

JSON is simple, easy to read, and works with any programming language. It's light - weighted scripts and travels fast over the internet, so it's used everywhere for API data.

21. One key difference between a Python dictionary and JSON.

A Python dictionary only exists inside a running Python program.

JSON is plain text used to send data between different systems over a network.

22. Convert to JSON: {"name": "Employee", "department": "IT", "active": True}

```
{  
  "name": "Employee",  
  "department": "IT",  
  "active": true  
}
```

23. Which function to (a) read, (b) create, (c) delete data?

To read - requests.get()

To create - requests.post()

To delete - requests.delete()

24. Purpose of the timeout parameter?

Timeout is used to stop the program from waiting forever if the server is slow or not responding or dead. Without it, the app could freeze.

25. Write a try/except block for calling an API safely.

```
try:
    response = requests.get(url, timeout=10)
    response.raise_for_status()
    data = response.json()

except requests.exceptions.RequestException as e:
    print("API call failed:", e)
```

26. Advantage of requests.Session() over calling requests.get() repeatedly?

A Session reuses the same connection for multiple requests, making things faster instead of opening a new connection every time.

27. Steps of how a chatbot prompt becomes a response.

User types a prompt.

It goes to a Python program.

The program sends it as an HTTP POST request to the AI API server.

The AI server sends it to the LLM model.

The model creates a response in JSON format.

The program extracts the text from it.

The user sees the final answer.

28. Four best practices when working with company APIs.

Never expose API keys — use .env files.

Always use HTTPS.

Set timeouts and handle errors properly.

Respect rate limits and read the documentation first.

29. Risk of hard-coding an API key and pushing it to GitHub, and the fix.

The key becomes public and anyone can misuse it, even after deleting it, since GitHub keeps history. The correct way is to store it in a .env file and keep it out of GitHub using .gitignore.

30. Summarize the full journey of an API call.

The client builds a request with a URL, method, headers, and sometimes a body. The request travels through the ISP, routers, and DNS to reach the server. The server checks

authentication, processes the request, and sends back a response with a status code and JSON data (or an error). The client then reads this response and shows the result to the user.